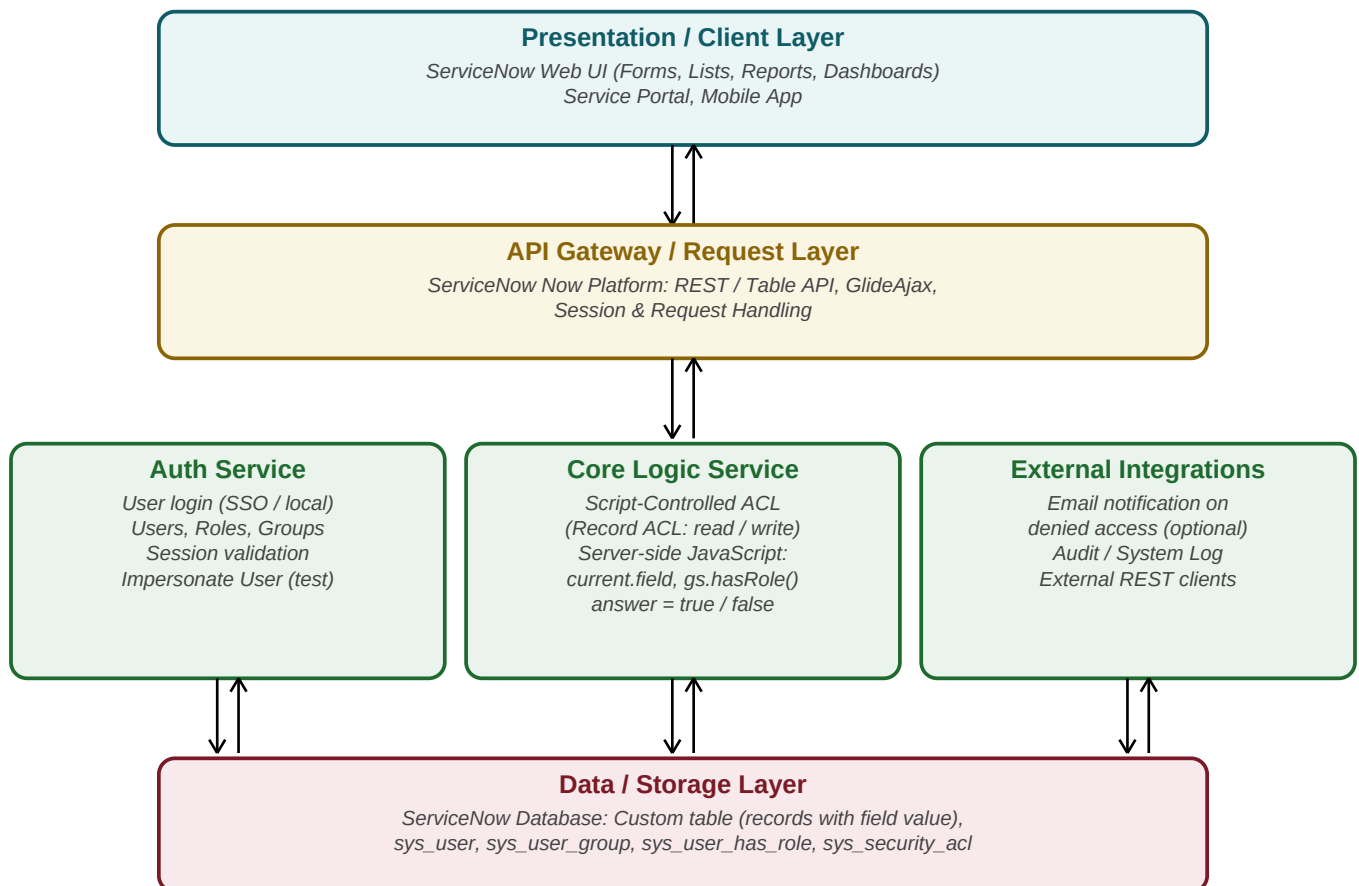


Solution Architecture

Date	29 September 2026
Team ID	SWTID-2026-1307
Project Name	Script-Controlled ACL-Restrict Record Access Based on Field Value.
Maximum Marks	5 Marks

Solution Architecture Diagram:

A high-level view of the solution: a user request passes through the platform layers, and the script-controlled ACL in the Core Logic layer compares the record's field value with the user's identity before the record is returned from the database.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Where users open, search and list records. Shows only the records and fields that the ACL allows for the logged-in user.	ServiceNow Web UI (forms, lists, reports), Service Portal, Mobile App
API Gateway	Receives every request (UI, list, report, search, REST) and passes it to the platform security check, so access cannot be bypassed from any channel.	ServiceNow REST / Table API, Now Platform request handling
Auth Service	Identifies the user and provides role, group and department details used by the ACL. Impersonate User is used to test different roles.	ServiceNow users, roles, groups, SSO / local login, sessions
Core Logic Service (Microservice)	Script-controlled ACL. Script reads current.<field> and compares it with the user (role, user ID, group). Sets answer = true to allow or false to deny, on the server side.	ACL (sys_security_acl), Server-side JavaScript, GlideRecord, gs.hasRole(), gs.getUserID()
External Integrations	Optional. Sends email notification or logs when access is denied, and supports audit review.	ServiceNow Email notifications, System Log / Audit, REST clients
Database	Stores the records with the sensitive field value, and the users, groups, roles and ACL definitions used for the decision.	ServiceNow Database: custom table, sys_user, sys_user_group, sys_user_has_role